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TECHNICAL DEPLOYMENT & DEVELOPER HANDOVER MANUAL

*An Operational Guide to Continuous Deployment, Monorepo Architecture,
DNS Governance, Google SMTP Secrets, and CMS Administration*

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Notice of Confidentiality & Executive Abstract

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Abstract

This technical manual serves as the comprehensive operational handover guide for the official corporate web platform of **THE HINTER GROUP GHANA LTD (HGG)**, accessible globally at <https://hintergroupghana.com>. Authored under a direct corporate infrastructure ownership framework, this document provides complete, step-by-step engineering protocols enabling HGG to govern, deploy, and maintain its digital infrastructure independently without ongoing reliance on third-party developer accounts.

The manual details the end-to-end architecture integrating a high-performance Next.js presentation engine deployed to Netlify's globally distributed Edge Network, coupled with a headless Sanity CMS content repository, Google SMTP mail routing, an enterprise Personnel Identity Verification engine, and native Search Engine Optimization (SEO) with Google Search Console verification. Clear, sequenced operational procedures encompass Git repository linking, monorepo workspace configuration (Development/frontend), production environment secrets injection, Google Workspace application password provisioning, SSL/TLS certificate enforcement, DNS TXT search verification, and comprehensive post-handover validation audits.

Core Engineering Capabilities Handed Over

- Complete source code, schemas, and assets maintained in the official GitHub repository (hintergroupdev/HGG).
- Continuous deployment pipeline connected directly to HGG's native Netlify hosting instance.
- Decoupled, serverless contact dispatch infrastructure powered by Google SMTP.
- Real-time content administration via Sanity Studio and QR-linked Personnel Verification.
- Enterprise on-page SEO generation and Google Search Console DNS TXT verification.

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Executive Architecture & Handover Strategy

1.1 System Overview

The official corporate web platform for **THE HINTER GROUP GHANA LTD (HGG)** has been constructed as an enterprise, high-performance static and serverless application. The architecture decouples the continuous integration and edge hosting infrastructure (Netlify) from the Headless Content Management System (Sanity.io) and corporate mail dispatch services (Google SMTP).

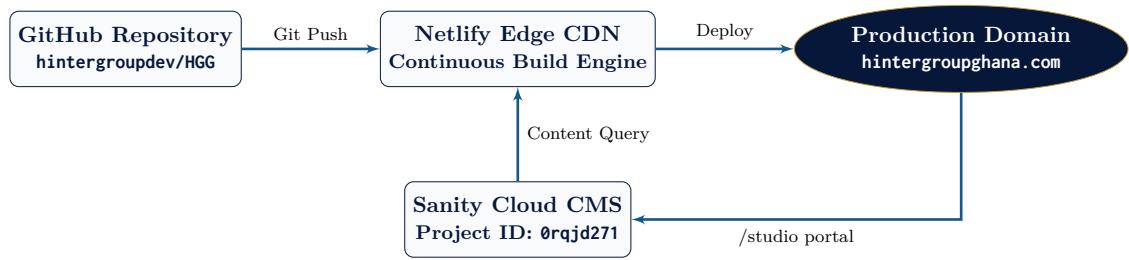


Figure 1.1: HGG Enterprise Platform Architecture Diagram

1.2 Direct Infrastructure Ownership Model

Under direct infrastructure ownership, HGG retains 100% legal, administrative, and technical control of its production Netlify environment. This eliminates the necessity of purchasing additional third-party team seats while ensuring your organization can independently trigger continuous deployments, rotate security keys, and manage domain records.

Master Credentials & Repository Registry

2.1 Master Access Registry

All code, graphical assets, and deployment templates are preserved in the official repository. Table 2.1 documents the master access parameters.

Resource Item	Identifier / URL	Operational Context
GitHub Repository	https://github.com/hintergroupdev/HGG	Complete source code, schemas, and configurations.
GitHub Account / Org	hintergroupdev	Authenticated exclusively via Google Single Sign-On (SSO).
Google Master Email	hintergroup.dev@gmail.com	Master gateway email for GitHub, Netlify, and Sanity.
Google Account Password	hintergroup1234	Master Google account password (gateway credentials).
Production Branch	main	Production-ready branch tracking live releases.
Production URL	https://hintergroupghana.com	Permanent live website domain.
Sanity Studio URL	https://hintergroupghana.com/studio	Embedded content management portal.
Verification URL	https://hintergroupghana.com/verify/HGG-001	Personnel identity verification route.

Table 2.1: Official Project Credentials and Master Endpoints

Authentication Architecture: Google Single Sign-On (SSO)

Crucial Note on Credentials: hintergroupdev and hintergroup1234 are **not** standalone GitHub username and password credentials. GitHub does not have a native password for this account.

- In all portals (GitHub, Netlify, Sanity Studio), access is authenticated by selecting **“Sign in with GitHub”**.
- Whenever no active GitHub session exists, GitHub requires selecting **“Sign in with Google”** using **hintergroup.dev@gmail.com** and password **hintergroup1234**.

2.2 Initial Login & Repository Access Protocol

Before linking Netlify or modifying environment variables, the client administration should authenticate into the official accounts in the following required sequence:

Step 1: Log In to the Project Email First: Open a web browser and navigate to Google / Gmail:

<https://mail.google.com>

Sign in with the project developer account:

- **Email Address:** hintergroup.dev@gmail.com

- **Google Password:** hintergroup1234

Operational Note: Keep this Gmail inbox open in your browser tab. When signing in to GitHub from a new computer, IP address, or location, security verification codes will arrive here.

Step 2: Sign In to GitHub via Google SSO: In a new browser tab, navigate to the GitHub authentication portal:

<https://github.com/login>

Important: Do not enter hintergroupdev or hintergroup1234 into GitHub's native login fields. The account has no independent GitHub password and is connected via Google Single Sign-On (SSO). Follow these steps:

- Click **Sign in with Google** (or *Continue with Google*).
- On the Google account selection screen, select the active project account: **hintergroup.dev@gmail.com** (already signed in from Step 1).
- GitHub will verify the Google session and grant access to **hintergroupdev**.
- If GitHub displays a one-time device verification screen, switch to the open Gmail tab, copy the 6-digit code received from GitHub, and enter it to complete sign-in.

Step 3: Navigate Directly to the Master Repository: Once successfully authenticated, visit the HGG production repository:

<https://github.com/hintergroupdev/HGG>

Confirm that the active branch is set to **main** and verify that you possess full administrative ownership over the codebase.

Password Hygiene Recommendation

Following successful production deployment, it is advised to log into <https://github.com> and <https://myaccount.google.com> as hintergroup.dev@gmail.com to update account passwords to your internal corporate standard.

2.3 Repository Monorepo Layout

The repository utilizes a modular structure where the production website application resides in the dedicated workspace directory Development/frontend:

```
HGG/                                <-- Repository Root
+-- netlify.toml                    <-- Root build & security headers
+-- Development/
  |-- frontend/                     <-- CRITICAL: NEXT.JS APP ROOT
    +-- app/                        <-- App Router (Pages, Verification, Studio)
      | +-- page.js                  <-- Homepage
      | +-- about-us/                <-- About Us (Foundation & Focus Areas)
      | +-- expertise/               <-- Service Pillars & Disciplines
      | +-- leadership/              <-- Executive Profiles
      | +-- verify/[id]/             <-- Personnel ID Verification & QR
      | +-- studio/                  <-- Sanity CMS Web Portal
      | +-- api/contact/             <-- SMTP Contact Form Dispatcher
    +-- components/                 <-- Reusable UI, Navbars, Footers
    +-- sanity/                     <-- CMS Schemas & Data Queries
```

```
+-- public/assets/      <-- Vector Logos, Favicons, QR Art
+-- package.json        <-- Node.js Dependencies & Scripts
\-- netlify.toml        <-- App-level Security & Routing
```

Netlify Re-Linking Protocol

3.1 Preserving the Existing Project & Domain

Because HGG already possesses an active Netlify site with `hintergroupghana.com` configured, **do not delete the project or create a new site**. Re-linking the repository directly inside the existing project preserves your custom domain, Let's Encrypt SSL certificates, and DNS records.

3.2 Step-by-Step Re-linking Sequence

Follow these 4 brief steps inside your Netlify dashboard:

Step 1: Open Your Active Site in Netlify: Sign in at <https://app.netlify.com> (select **Log in with GitHub**; with your active browser session from Chapter 2, Netlify authenticates automatically), go to the **Sites** tab, and click on your active production site (<https://hintergroupghana.com>).

Step 2: Navigate to Continuous Deployment: In the left sidebar, click **Site configuration** → select **Build & deploy** → open **Continuous deployment**.

Step 3: Connect to GitHub: In the **Repository** settings card:

- If an older repository is currently attached: Click **Manage repository** → select **Link to a different repository** (or **Unlink repository**).
- If no repository is linked: Click **Link repository**.
- In the Git provider modal, choose **GitHub**.

Step 4: Authorize & Select Repository: When the GitHub authorization window appears:

- Authorize Netlify under the `hintergroupdev` account (or select organization `hintergroupdev`).
- Search and select the master repository: **hintergroupdev/HGG**.
- Ensure the deployment branch is set to **main**.
- Click the primary **Deploy site** (or **Deploy / Save**) button to confirm the repository connection.

Build Settings Specification

4.1 Mandatory Monorepo Paths

In your Netlify dashboard, navigate to **Site configuration** → **Build & deploy** → **Continuous deployment**, locate the **Build settings** section, click **Edit settings**, and enter the following values:

Setting Field	Exact Value to Enter	Operational Context
Base directory	Development/frontend	Directs Netlify into the Next.js application workspace.
Build command	npm run build	Compiles optimized SSR bundles and pages.
Publish directory	.next	Build artifact directory (relative to Base directory).
Branch to deploy	main	Master branch tracking stable production releases.

Table 4.1: Required Netlify Build Configuration Fields

Notice on Publish Directory

Because the **Base directory** is already set to `Development/frontend`, Netlify automatically resolves all paths from inside that subfolder. Enter **.next** only as the Publish directory. Do not write `Development/frontend/.next`.

Action: Triggering the Deployment

Once the parameters above are entered, click the primary **Deploy site** (or **Deploy / Save**) button at the bottom of the settings card to trigger the continuous build process.

Environment Variables & Secrets (.env)

5.1 Configuration Block & Mandatory Modifications

In Netlify, navigate to **Site configuration** → **Environment variables**, click **Add a variable** → select **Import from a .env file**, and paste the block below.

Critical Instruction: Modify Placeholders Before Saving

Do NOT save with raw placeholder values. You may paste the configuration block below, but you **must modify the 3 email credential values** before clicking Save:

- **SMTP_USER:** Replace with your sender Google email (hintergroup.dev@gmail.com).
- **SMTP_PASS:** Replace with the 16-character Google App Password (Chapter 6).
- **CONTACT_RECEIVER_EMAIL:** Replace with your corporate receiving inbox.

```
NEXT_PUBLIC_SITE_URL=https://hintergroupghana.com
NEXT_PUBLIC_SANITY_PROJECT_ID=0rqjd271
NEXT_PUBLIC_SANITY_DATASET=production
NEXT_PUBLIC_SANITY_API_VERSION=2024-08-30
SANITY_API_READ_TOKEN=
SMTP_HOST=smtp.gmail.com
SMTP_PORT=465
SMTP_SECURE=true
SMTP_USER=YOUR_SENDER_GMAIL_HERE
SMTP_PASS=YOUR_16_DIGIT_APP_PASSWORD_HERE
CONTACT_RECEIVER_EMAIL=YOUR_RECEIVING_INBOX_HERE
```

5.2 Parameter Explanations

Variable Key	Value / Source	Operational Role
NEXT_PUBLIC_SITE_URL	https://hintergroupghana.com	Canonical URL for SEO and QR verification.
NEXT_PUBLIC_SANITY_PROJECT_ID	0rqjd271	HGG's Sanity database identifier.
NEXT_PUBLIC_SANITY_DATASET	production	Primary live data store.
NEXT_PUBLIC_SANITY_API_VERSION	2024-08-30	Fixed API version lock.
SANITY_API_READ_TOKEN	(Leave Empty)	Public read permissions enabled; token not required.
SMTP_HOST	smtp.gmail.com	Google SMTP server for contact form dispatch.
SMTP_PORT	465	Secure SSL port.
SMTP_SECURE	true	Enforces SSL socket encryption.
SMTP_USER	Sender Google Address	Email address utilized to transmit contact notifications.
SMTP_PASS	16-Digit App Password	Dedicated Google Application Password (Chapter 6).
CONTACT_RECEIVER_EMAIL	Recipient Inbox	Destination corporate inbox (e.g. info@...).

Table 5.1: Environment Variables Specification

Google SMTP & App Password Setup

6.1 Security Rationale

To protect corporate accounts against credential stuffing and brute-force intrusion, Google prohibits automated web servers from authenticating using primary account passwords. Instead, Google requires a **16-character dedicated Application Password**.

Prerequisite

Google requires **2-Step Verification** to be active on the sending account before an Application Password can be generated.

6.2 Step-by-Step Generation Protocol

Step 1: Access Account Security: Log into the Google/Gmail account you designated for SMTP_USER (such as hintergroup.dev@gmail.com or your corporate Google Workspace address) and visit:

<https://myaccount.google.com/security>

Step 2: Verify 2-Step Verification Status: Under the *"How you sign in to Google"* section, ensure **2-Step Verification** is set to **On**. If not, complete the prompt using your mobile device.

Step 3: Generate the 16-Character Password:

- In the search box at the top of the Google Account portal, type: **App passwords** and select it.
- In the **App name** field, enter: HGG Website Contact Form.
- Click **Create**.
- Google will display a popup containing a **16-character code** (formatted like xxxx xxxx xxxx xxxx).
- Copy this 16-character string immediately.

Step 4: Save in Netlify:

- Return to Netlify → **Site configuration** → **Environment variables**.
- Set SMTP_PASS to your 16-character code.
- Set SMTP_USER to your Google address.
- Set CONTACT_RECEIVER_EMAIL to your corporate inquiry inbox.
- Click **Save**.

Production Deployment & SSL Verification

7.1 Triggering the Deploy

With the repository linked and secrets configured, you can launch the live build:

1. In the Netlify dashboard, click the **Deploys** tab in the top navigation.
2. Click **Trigger deploy** → select **Deploy site** (or **Clear cache and deploy site**).
3. Click on the active deploy to view the streaming build terminal.
4. The build sequence will complete within 2 minutes with:

Site deploy was successfully published!

7.2 Domain & SSL Health Check

1. Go to **Site configuration** → **Domain management**.
2. Verify that `hintergroupghana.com` displays with a green checkmark as the **Primary domain**.
3. Scroll to **HTTPS**: verify that an active **Let's Encrypt SSL certificate** is issued. If not, click **Verify DNS configuration** → **Provision certificate**.

Sanity Studio CMS Administration

8.1 Accessing the Live CMS Portal

The Content Management Studio is embedded natively within the production website:

<https://hintergroupghana.com/studio>

8.2 Signing In to Sanity Studio

Sanity Studio authentication is federated through GitHub. Follow these exact steps:

1. On the Sanity Studio login screen, click **Sign in with GitHub**.
2. **If you already have an active GitHub session** in your browser (from Chapter 2):
Sanity Studio will immediately authenticate and grant you access to the HGG Content Studio dashboard.
3. **If no active GitHub session exists (e.g., in a private or new browser window):**
 - The GitHub login portal will appear.
 - **Do NOT attempt to type hintergroupdev or hintergroup1234 into the GitHub username/password fields** (they are not standalone GitHub credentials).
 - Click **Sign in with Google** (or *Continue with Google*).
 - Choose or enter **hintergroup.dev@gmail.com** and password **hintergroup1234**.
 - If prompted, click **Authorize Sanity.io**.
4. You will enter the HGG Content Studio with full administrator privileges.

8.3 Inviting Colleagues & Assigning Editorial Roles

To grant access to executives or HR staff using their personal or corporate emails without sharing GitHub credentials:

1. Visit the Sanity Management Console: <https://www.sanity.io/manage>.
2. Click **Log in with GitHub**. (If GitHub prompts for login, select **Sign in with Google** using `hintergroup.dev@gmail.com`).
3. Select project **HGG** (Project ID: `0rqjd271`).
4. Click the **Members** tab in the top navigation bar.
5. Click **Invite Member**.
6. Enter the colleague's email address and assign an appropriate role:
 - **Administrator:** Complete control over schema, dataset, user permissions, and billing.
 - **Editor:** Permission to update site text, post articles, and issue employee credentials.
7. Click **Send Invite**. The recipient will receive an activation email and can subsequently log in directly at <https://hintergroupghana.com/studio>.

Search Engine Optimization & Google Search Console

9.1 Native Enterprise SEO Architecture

The HGG corporate web platform incorporates automated, enterprise-grade search engine optimization natively within the application code. Every page is pre-rendered with complete metadata, semantic markup, and crawler assets:

- **Dynamic Metadata Engine (`app/layout.js`):** Employs the Next.js Metadata API to dynamically construct page titles using the corporate format (%s | HGG), tailored meta descriptions, targeted keywords, author and publisher attribution, and canonical link tags enforcing <https://hintergroupghana.com> to prevent duplicate indexing.
- **Social Graph Previews (OpenGraph & Twitter Cards):** Full OpenGraph (`og:title`, `og:description`, `og:image`, `og:url`) and Twitter Card metadata (`summary_large_image`) are generated automatically. High-resolution 1200 × 630px preview cards display seamlessly whenever links are shared on LinkedIn, WhatsApp, X, or corporate messengers.
- **Structured Data (Schema.org JSON-LD):** Injected via `components/seo/JsonLd.jsx` across every page, defining HGG as an official Corporation / Organization entity. Microdata fields register legal entity naming, official logo endpoints, headquarters location (Accra, Ghana), contact points, and verified URLs to qualify HGG for Google Knowledge Graph recognition.
- **Automated XML Sitemap (`sitemap.xml`):** Programmed dynamically in `app/sitemap.js` and accessible at <https://hintergroupghana.com/sitemap.xml>. Automatically indexes all static marketing routes alongside every dynamic Insights article published via Sanity CMS, assigning respective `lastModified` timestamps, `changeFrequency`, and `crawl priority` weights.
- **Search Engine Crawl Directives (`robots.txt`):** Configured in `app/robots.js` and accessible at <https://hintergroupghana.com/robots.txt>. Provides explicit crawl directives for Googlebot, Bingbot, and Applebot. Allows public indexing of corporate content while securely disallowing internal administrative routes (`/studio/`, `/api/`, `/verify/`), and directly references the live XML sitemap.
- **Web App Manifest (`app/manifest.js`):** Provides Progressive Web App (PWA) manifest compliance, enabling mobile browsers to index standalone app capabilities and icons.

9.2 The Strategic Purpose of Google Search Console

While the codebase automatically generates all technical SEO signals, search engines require formal domain ownership verification to activate search indexing. Google Search Console (GSC) is Google's official webmaster platform, serving critical operational functions:

1. **Search Indexing Activation:** Explicitly informs Googlebot of the live domain, initiating scheduled indexing across worldwide Google Search results.
2. **Direct Sitemap Ingestion:** Directly ingests `sitemap.xml` so new corporate pages and Sanity CMS news publications are indexed rapidly.

- 3. **Search Performance Analytics:** Tracks organic impressions, search queries, click-through rates (CTR), and average ranking positions for HGG brand keywords.
- 4. **Health & Security Monitoring:** Automatically alerts the organization to broken links, mobile usability faults, Core Web Vitals performance, and security notifications.

9.3 DNS TXT Verification Protocol

Google provides several verification mechanisms. The **DNS TXT Record method via the Domain property** is the industry standard and the required method for HGG, as it universally verifies all subdomains (hintergroupghana.com and www.hintergroupghana.com) and all protocols (http and https) simultaneously.

Step 1: Access Google Search Console: Open a web browser and navigate to the Google Search Console portal:

<https://search.google.com/search-console>

Sign in using the master corporate Google account:
hintergroup.dev@gmail.com.

Step 2: Select the Domain Property Type: On the **Select property type** modal screen:

- Choose the **Domain** option (left card), *not* the “URL prefix” card.
- Enter the root domain name exactly without prefixes or slashes:

hintergroupghana.com

- Click **Continue**.

Step 3: Copy the Unique DNS TXT Verification String: Google will open a dialog window entitled **Verify domain ownership via DNS record**:

- In the **Record type** dropdown, ensure **TXT** is selected (recommended).
- Google displays a unique verification key formatted as:
google-site-verification=XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
- Click the **Copy** button to save this complete string to your clipboard.

Step 4: Add the TXT Record to Your DNS Provider: Log in to the DNS provider dashboard managing the nameservers for hintergroupghana.com (e.g., Netlify DNS, Cloudflare, Namecheap, or your corporate domain registrar).

- Navigate to **DNS Management** (or **DNS Settings / Zone Editor**).
- Click **Add new record** (or *Add record*).
- Configure the new record parameters as specified in Table 9.1:

DNS Record Field	Value to Enter	Operational Purpose
Record Type	TXT	Text verification record.
Name / Host	@ (or hintergroupghana.com)	Root apex domain pointer.
Value / Content	<i>Paste Google TXT string</i>	google-site-verification=...
TTL	3600 (or 1 hour / Auto)	Time-To-Live DNS cache duration.

Table 9.1: DNS TXT Record Configuration for Google Search Console

- Click **Save** (or *Create Record*).

Step 5: Confirm Verification in Google Search Console: Return to the open Google Search Console browser tab and click the green **Verify** button.

- *Propagation Note:* DNS updates typically propagate across global recursive resolvers within 2 to 15 minutes. If Google initially reports that the verification token was not found, wait 5 minutes and click **Verify** again.
- When verification succeeds, Google will display a confirmation dialog: **Ownership verified**.
- Click **Go to property** to open your new domain dashboard.

Step 6: Submit the Production XML Sitemap: To initiate immediate web crawler indexing:

- In the left-hand navigation sidebar, click **Sitemaps** (under the **Indexing** group).
- In the **Add a new sitemap** text box, enter:

sitemap.xml

(*Note:* The domain prefix <https://hintergroupghana.com/> is automatically prepended by Google).

- Click **Submit**.
- Google Search Console will process the request and update the status column to **Success**. Googlebot will now continuously ingest and refresh all website pages and CMS publications.

Quality Assurance & Verification Protocols

Following deployment, execute the following five validation checks:

1. Personnel Verification System & QR Audit:

- Open: <https://hintergroupghana.com/verify/HGG-001>
- Verify the green shield badge displays **IDENTITY VERIFIED**.
- Verify the live clock is active and synchronized to **Accra, Ghana (GMT)** time.
- Verify that strictly public professional credentials appear (Photo, Name, Title, Org, Status, Issue Date).
- Verify that **no confidential HR data** (DOB, home address, personal phone/email, national ID) appears on the page.
- Scan the physical QR code on the sample ID card using a smartphone camera (iOS / Android); it must open directly to <https://hintergroupghana.com/verify/HGG-001>.

2. Contact Desk Email Routing Test:

- Open: <https://hintergroupghana.com/contact>
- Submit a test inquiry with your name and email.
- Check the inbox defined in CONTACT_RECEIVER_EMAIL to verify prompt delivery.

3. CMS Live Synchronization Test:

- Log into <https://hintergroupghana.com/studio>.
- Make a minor test edit (e.g. updating office phone or an article excerpt) and click **Publish**.
- Refresh the public site to confirm the update reflects immediately.

4. Multi-Device Responsive Audit:

- Verify layout on Desktop (Chrome, Safari, Edge).
- Verify layout on Tablet (iPad, Android Tablet).
- Verify layout on Mobile (iPhone iOS, Android).
- Ensure mobile hamburger navigation opens smoothly and no horizontal scroll occurs.

5. Search Engine Optimization & Sitemap Audit:

- Open: <https://hintergroupghana.com/sitemap.xml> in your browser; verify that all corporate routes and insights articles render cleanly in valid XML.
- Open: <https://hintergroupghana.com/robots.txt>; verify that search engine crawlers (Googlebot, Bingbot, Applebot) are granted access, while administrative routes are protected.
- Check Google Search Console to confirm the Domain property displays verified status and the sitemap reports **Success**.

Operational Continuity & Support SLA

11.1 Automated Git Workflow

Any future code updates, design enhancements, or schema additions pushed to the `main` branch of <https://github.com/hintergroupdev/HGG> will automatically trigger a Netlify production build. Zero manual intervention is required.

11.2 Technical Support Protocol

If you or your team experience any difficulty during this deployment:

1. Note the specific chapter and step number where the issue arose.
2. Capture a screenshot of the Netlify build terminal or error dialog.
3. Contact me directly, and I will assist you until the deployment is completed.